Mode choice of Swiss Metro

Course Project

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by

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Abstract

This study investigates the mode choice behavior of individuals in Switzerland regarding the hypothetical Swiss Metro (SM), train, and car, employing advanced discrete choice models to capture heterogeneity and derive policy insights. Using the Swiss Metro dataset, we estimate three models—Multinomial Logit (MNL), Nested Logit (NL), and Mixed Multinomial Logit (MMNL)—to address the limitations of traditional approaches. The MNL model confirms the negative impact of travel time and cost on utility but is constrained by the Independence of Irrelevant Alternatives (IIA) property. The NL model reveals stronger substitution patterns between train and car, suggesting shared unobserved characteristics. The MMNL model, incorporating random taste heterogeneity, identifies the log-normal distribution as the best fit for travel time and cost, highlighting significant preference variations across age and income groups. Policy simulations indicate that reducing train costs or headways can shift demand from SM to trains, alleviating congestion. Willingness-to-pay (WTP) analysis, though challenging under log-normal assumptions, yields a plausible estimate of 0.83 Swiss Francs per minute saved using uniform distribution. The study underscores the importance of accounting for heterogeneity in transportation planning and suggests future research directions.

Keywords: Mode choice, Swiss Metro, Discrete choice models, Mixed Multinomial Logit, Heterogeneity, Willingness-to-pay.

Chapter 1

Introduction

This research revisits the Swissmetro dataset originally analyzed by (Bierlaire *et al.* 2001), which explores individual mode choice behavior between Train, Car, and a hypothetical high-speed Swissmetro option.
The original study employed models from the Generalized Extreme Value (GEV) family—such as the Multinomial Logit (MNL), Nested Logit (NL), and Cross-Nested Logit (CNL)—to demonstrate how capturing the correlation structure between alternatives and error components can significantly improve behavioral insights. Building upon this foundational work, our study proposes a more nuanced and modern exploration of the same dataset by implementing three discrete choice models: the standard MNL, the NL and the Mixed Multinomial Logit (MMNL), to capture intra-individual heterogeneity by allowing random variation in preferences across individuals.
These models allow us to account for random taste variation, nesting structure, and unobserved population heterogeneity, respectively.
The remainder of this paper is organized as follows: Chapter 2 reviews basic and advanced models in domain of econometric choice modeling touching briefly on SP and RP data collection methods. Chapter 3 describes the dataset used for the model. Chapter 4 identifies the research gap from existing literature and presents the paper's objectives. Chapters 5 and Chapter 6 detail the methodology and model estimation results, respectively. Chapter 7 discusses policy analysis based on the dataset. Chapter **??** highlights the limitations and future scope of the study. Finally conclusion are drawn in Chapter 8

Chapter 2

Theoretical Framework & Literature Review

2.1 Introduction to RUM and the GEV Framework

The field of econometric modeling has undergone a significant transformation, with Random Utility Maximization (RUM) emerging as a dominant framework for modeling individual choice behavior. This framework has become especially influential in fields such as transport, marketing, and environmental economics. Over the past few decades, there has been a sharp rise in both theoretical exploration and empirical application of RUM-based models.
As (Bhat 1997) noted, this advancement can be attributed to two main factors , first the development of new class models within the Generalized Extreme Value (GEV) framework, and second, significant progress in simulation methods that allow for the estimation of more complex choice models.

The introduction of new GEV models has provided more flexibility in capturing realistic behavioral structures while still maintaining an analytically tractable closed-form choice probability and likelihood function. Additionally, advances in simulation techniques have facilitated the estimation of choice models that would otherwise be mathematically intractable, thus broadening the scope of econometric modeling beyond simpler models.

These advancements in model specification have led to the introduction of more complex and flexible models, particularly within the GEV framework. These models overcome the limitations of earlier models by allowing for greater behavioral realism. However, before diving into these newer, more complex models, it is essential to first understand the foundational model within the GEV framework: the Multinomial Logit (MNL) model. Understanding MNL will provide context for the limitations that led to the development of more advanced GEV models.

2.2 Multinomial Logit (MNL) Model and Its Limitations

The Multinomial Logit (MNL) model, introduced by (McFadden 1972), has been a cornerstone in discrete choice modeling due to its simplicity and closed-form solution for estimating choice probabilities. In the context of Random Utility Maximization (RUM), the MNL model assumes that individuals select the alternative that provides the highest utility, which is comprised of both a systematic (observable) component and a random (unobservable) component. The probability that individual n chooses alternative i from a set C is given by:

$$P_{ni} = \frac{\exp(V_{ni})}{\sum_{j \in C} \exp(V_{nj})}$$

Despite its computational simplicity and closed-form probability estimation, the MNL model has three key limitations (Ben-Akiva & Lerman 1985) at individual level

1. **Independence of Irrelevant Alternatives (IIA)**: The MNL model assumes that the relative odds of choosing between any two alternatives are unaffected by the presence of additional alternatives, which is often unrealistic. This assumption leads to constrained substitution patterns between alternatives.

2. **Homogeneity of Preferences**: The model assumes all individuals respond identically to the attributes of the alternatives. However, individuals may have different sensitivities to these attributes, and ignoring this variation can result in biased estimates.

3. **Homogeneous Error Structure**: The MNL model assumes that the variance of unobserved factors influencing utility is the same for all individuals and alternatives. In reality, this assumption is often violated, as different alternatives may carry different levels of uncertainty or variability for different individuals.

2.3 GEV Class Models

The foundational work on the Generalized Extreme Value (GEV) class of models was laid out by (McFadden 1977), who derived this class from the Random Utility Maximization (RUM) framework. In contrast to the Multinomial Logit (MNL) model, which assumes independently and identically distributed (i.i.d.) extreme value error terms across alternatives, the GEV formulation allows for a generalized extreme value structure. This permits correlation among the unobserved components of utility across alternatives.

McFadden introduced the concept of a *generating function*, denoted $G(y_1, y_2, \ldots, y_J)$, which governs the joint distribution of the unobserved utility components. The closed-form expression for the choice probability of alternative i is then given by:

$$P(i) = \frac{\frac{\partial G(y)}{\partial y_i}}{G(y)}, \quad \text{where } y_j = \exp(V_j), \quad \forall j \in \{1, 2, \ldots, J\}$$

where V_j is the deterministic utility component of alternative j. The function G must satisfy specific mathematical conditions such as homogeneity of degree one,

non-negativity, and the property that it approaches infinity as any $y_j \to \infty$. These conditions ensure that the resulting choice probabilities are consistent with utility maximization and lie between 0 and 1.

This theoretical contribution enabled the construction of a broader class of discrete choice models capable of representing flexible substitution patterns, which are not possible under the restrictive IIA property of the MNL model.

Building upon McFadden's theoretical foundation, (Ben-Akiva & Lerman 1985)developed practical estimation techniques for a subclass of GEV models known as Nested Logit (NL) models. The NL model introduces a two-level decision structure in which alternatives are grouped into nests, and correlation is permitted among alternatives within the same nest, but not across nests. This nesting structure is particularly useful in transportation modeling where alternatives often exhibit shared unobserved characteristics.

The generating function for the NL model takes the form:

$$G(y_1, \ldots, y_J) = \sum_{m \in \mathcal{M}} \left(\sum_{j \in \mathcal{C}_m} y_j^{1/\lambda_m} \right)^{\lambda_m}$$

where $\mathcal{C}_m$ represents the set of alternatives in nest m, and $\lambda_m \in (0, 1]$ is the dissimilarity parameter for that nest. The parameter λ_m captures the degree of independence among alternatives in the same nest, with lower values indicating stronger correlation.

2.4 Mixed Multinomial Logit (MMNL) Model

The Mixed Multinomial Logit (MMNL) model is an extension of the Multinomial Logit (MNL) model that allows for random variation in the coefficients across individuals.(McFadden & K. Train 2000)have demonstrated that, under certain conditions, a mixed logit model can approximate a wide range of choice models with a high degree of accuracy. In contrast to the fixed coefficients in MNL, MMNL assumes that the preferences (coefficients) of individuals can vary according to some underlying distribution. This structure allows for greater flexibility in modeling heterogeneity in decision-making (Greene & Hensher 2003) (Revelt & K. E. Train 1998)

The basic MNL model calculates the probability of choosing an alternative i as a function of the deterministic utility $V_{qi} = \beta' x_{qi}$ (where β represents the fixed parameters and x_{qi} the observed variables). The probability is given by:

$$P_{qi} = \frac{e^{V_{qi}}}{\sum_j e^{V_{qj}}}$$

However, in MMNL, the coefficients β are treated as random variables, and we integrate over their distribution, denoted by $f(\beta)$. This leads to the following expression for the probability of choosing alternative i:

$$P_{qi} = \int \frac{e^{\beta' x_{qi}}}{\sum_j e^{\beta' x_{qj}}} f(\beta) d\beta$$

Since this integral does not have a closed-form solution, simulation methods such as Monte Carlo simulation are used to approximate the choice probabilities by drawing random samples from the distribution of β (McFadden & K. Train 2000) (K. E. Train 2009).

2.5 Types of Data in Choice Modeling

In choice modeling experiments, the data collected can either be Stated Preference (SP) or Revealed Preference (RP) type. Although it is not always possible to obtain both types of data, having both formats is beneficial for consolidating results with empirical backing.

In the case of the SwissMetro dataset, as a novel mode of transit, only SP data is available. RP data, on the other hand, is based on real-world observations of actual choices made by respondents, reflecting their true behavior. The key advantage of RP data is that it provides insights into real-world decisions. However, it presents challenges such as limited variation in the available choices and the issue of multicollinearity, where some factors may be highly correlated, complicating econometric analysis (Louviere *et al.* 2000)

While SP data is not inherently superior to RP data, it offers unique advantages when only SP data is available. For instance, SP data allows for the simulation of hypothetical scenarios and policy changes, such as mobility pricing, which may not yet exist. This capability is valuable for exploring future market conditions and policy impacts. Additionally, SP data provides the ability to control the attributes included in the experiment, enabling the examination of non-existing market attributes that could be relevant in future contexts. This is the only way to assess scenarios where future or hypothetical conditions can be manipulated . Furthermore, SP data allows for controlling the correlation between included attributes, making the analysis more precise.(K. E. Train 2009)

Chapter 3

Study area and data description

3.1 Data Collection

Since SwissMetro was a futuristic and non-existing transport system at the time of study, traditional revealed preference (RP) data—based on actual traveler behavior—was not suitable for forecasting its potential usage. Instead, researchers used stated preference (SP) surveys, a well-established method in transport studies, to present people with hypothetical travel scenarios including SwissMetro as an option. This approach was adopted by (Bierlaire *et al.* 2001) to collect data from two main traveler groups: rail users and road users.

For rail users, surveys were conducted during actual train journeys between St. Gallen and Geneva in March 1998, with 470 passengers initially approached. Respondents provided details about their current trip—such as origin and destination—which were then used to create nine hypothetical travel choices tailored to their journey. Each choice included options for traditional rail, SwissMetro, and car (if the respondent owned one). The travel options were described by key factors like door-to-door travel time, fare, and waiting time (headway). For SwissMetro, additional features like seating class were specified to reflect comfort. Travel times were estimated using realistic assumptions about speed and distance, and the SwissMetro fare was calculated as 1.2 times the current rail fare to reflect faster service. After cleaning the data, 435 complete and usable responses were retained.

For car users, a household or phone-based survey was considered inefficient. Instead, license plates were recorded using cameras at major highways near Zürich, Bern, and Geneva in September 1997, collecting 10,529 plates. The central Swiss vehicle registry then sent survey packs to car owners, asking about their recent trips and interest in a follow-up survey. By April 1998, around 1,758 people responded, out of which 1,070 completed the full survey. A second SP survey was then sent to these respondents, using the same method as with rail users. Finally, 770 usable SP responses from road users were added to the dataset. Despite the gap between the trip and the survey, the 18 % response rate for the first mailing was considered strong, and the 72 % response rate for the follow-up SP survey was well within expectations.



Figure 3.1: Proposed Route of Swiss-Metro

3.2 Variables & Descriptive Statistics

Initially, we had observations for 1192 individuals, with each individual having 9 observations. However, for one individual, the chosen mode was not recorded, and thus we removed all 9 of that individual's observations. This left us with observations for 1191 individuals.

Again, among those 1191 individuals, some of them were carrying Swiss annual ticket and due to that their ticket number was given given in place of the cost of train and metro mode which is usually in higher digits as compared to actual costs. While, we were running our basic model (MNL) without removing their observations, due to that the co-efficient of cost was turning out to be positive which is not at all consistent..

Thus, we also removed the observations of those 169 individuals, leaving us with 1022 individual observations.
Again, regarding the income category, there were some individuals who had not revealed their income, and it is difficult to perform the analysis with a dataset where some respondents did not share their income. Thus, we removed another 108 individuals' observations, and in our final dataset, we were left with only 914 individual observations, with each individual having 9 observations..

Now, let's see the data description of those 914 individuals. We will divide our data data description into two categories and these are

- Description for descriptive variables (shown in table 3.1 & 3.2), and
- Description for numeric variables (shown in table 3.3)

Table 3.1: Data description for descriptive variables

Variables	Numeric values in the dataset	Data description	Percentage of population in the dataset used for analysis
Travel purpose	1	Commuter	14.11%
	2	Shopping	8.42%
	3	Business	49.12%
	4	Leisure	25.82%
	5	Return from work	0.88%
	6	Return from shopping	0.44%
	7	Return from business	0.88%
	8	Return from leisure	0.11%
	9	Other	0.22%
Travel ticket	1	Two way with half price card	50.88%
	2	One way with half price card	1.75%
	3	Two way normal price	37.75%
	4	One way normal price	0.98%
	5	Half day	2.19%
	6	Annual season ticket	1.75%
	7	Annual season ticket Junior or Senior	0.33%
	8	Free travel after 7pm card	2.95%
	10	Other	1.42%
Luggage	0	None	41.8%
	1	One piece	56.6%
	3	Several pieces	1.6%
Age	1	Age ≤ 24	3.61%
	2	$24 <$ Age ≤ 39	30.53%
	3	$39 <$ Age ≤ 54	38.84%
	4	$54 <$ Age ≤ 65	20.24%
	5	Age above 65	6.78%
Income	1	Below 50 thousands CHF	14.55%
	2	Between 50 to 100 thousands CHF	41.03%
	3	Above 100 thousands CHF	44.42%
Gender	0	Female	21.55%
	1	Male	78.45%
Metro availability	1	Available	100%
Train availability	1	Available	100%
Car availability	0	Not available	8.92%
	1	Available	91.08%

Table 3.2: Data description for descriptive variables

Variables	Numeric values in the dataset	Data description	Percentage of population in the dataset used for analysis
Choice indicator	1	Chosen mode is train	7.80%
	2	Chosen mode is swiss metro	58.40%
	3	Chosen mode is car	33.80%
Survey performed	0	Performed in train	24.84%
	1	performed in car	75.16%
Seats configuration in swiss metro	0	No airline seats	91.80%
	1	Airline seats	8.20%

Table 3.3: Data description for numeric variables

Variables	Mean	Standard deviation	Min value	Max value
Train travel time (in minutes)	176.04	78.89	31	1049
Train cost (in CHF)	98.13	64.65	4	576
Train headway (in minutes)	69.96	37.41	30	120
Swiss metro travel time (in minutes)	92.84	56.56	8	796
Swiss metro cost (in CHF)	118.9	80.03	6	768
Swiss metro headway (in minutes)	20.06	8.17	10	30
Car travel time (in minutes)	139.46	86.93	0	1560
Car cost (in CHF)	87.82	52.13	0	312

3.3 Transformation of descriptive variables for using in the model

In our dataset, we have few categorical variables like income, luggage, age etc. which are better not to directly include into our model. Rather, if we transform them into different categories or sections, that will capture the better effect of the population. For that, we are also transforming the relevant variables into different categories as shown in the table 3.4.

Table 3.4: Data description for descriptive variables

Variables	Category	Range	Percentage of population in the dataset used for analysis
Income	Low income (LI)	$<$ 50 thousand CHF	14.55%
	Medium income (MI)	Between 50 to 100 thousand CHF	41.03%
	High income (HI)	Above 100 thousand CHF	44.42%
Luggage	No luggage	0	41.8%
	Luggage	>0	58.2%
Age	Young age	Age $\leq$ 24	3.61%
	Mid age	24 $<$ Age $\leq$ 54	69.37%
	Old age	Above 54	27.02%

$\Rightarrow$ Though we have divided the income into three categories but after performing co-variance matrix analysis for all the variables as shown in the tables A.1 and A.2 , we can observe that there is high correlation between medium income and high income category. Thus, while performing our analysis we are treating them as same i.e. we are only having high and low income category (**keeping high income as base**).

Chapter 4

Research gap and objective of the study

Generalized Extreme Value (GEV) models assume that taste parameters are fixed across individuals, which simplifies the estimation process but fails to capture the heterogeneity in individual preferences. In contrast, the Mixed Multinomial Logit (MMNL) model allows for random taste heterogeneity, acknowledging that individuals may exhibit different preferences or trade-offs when choosing between alternatives.

This random heterogeneity in preferences is particularly important in scenarios such as mode choice prediction — for example, when individuals are deciding between alternatives like car, train, or flight. In these cases, the MMNL model typically outperforms the Multinomial Logit (MNL) model (McFadden & K. Train 2000), which assumes uniform preferences across all individuals. By accounting for this individual-level variation, the MMNL model provides more accurate and realistic predictions.

Previous studies on the swiss metro (**SM**) dataset have primarily focused on estimating mode choice using models such as MNL or MMNL. However, they have often overlooked the full consideration of heterogeneity in travel behavior, particularly the interaction between socio-demographic characteristics (e.g., age, income, and education) and travel attributes (e.g., travel time, cost, and convenience).(Bhat 1998)

This research addresses this gap by explicitly accounting for heterogeneity in travel behavior, providing a more nuanced understanding of how individual characteristics influence mode choice. By incorporating random taste variation in our models, we are able to better capture the diverse preferences of travelers and improve the accuracy of mode choice predictions.

Chapter 5

Methodological framework

In this chapter, we will discuss how we are formulating our study to align with our objectives

Our first objective is to capture the policy implications at base case and policy case. For that we will run two basic models named multinomial logit model (**MNL**) and nested logit model (**NL**). Basically we will perform the nested logit model to capture the drawbacks of IIA (independence of irrelevant alternatives) property and show the actual correlation among the alternatives for better policy implications.

Our another major objective was to find out the taste heterogeneity of the individuals towards travel time and travel cost. And it is always not necessary that both travel time and travel cost comes from the same distribution supporting the best fit of our data. For that we will randomize them based on different distributions and at the end pick the best fit distribution among them by using AIC BIC value as per (Ghosh *et al.* 2013). For that, we will use mixed multinomial logit model (**MMNL**). Consequently, we will also find out the willingness to pay of those individuals for the same distribution.

One of our major assumptions was that all 9 observations for an individual were independent of each other. To check this, we will randomize the constant terms.

Methodological framework can be understood better with the help of the diagram 5.1.

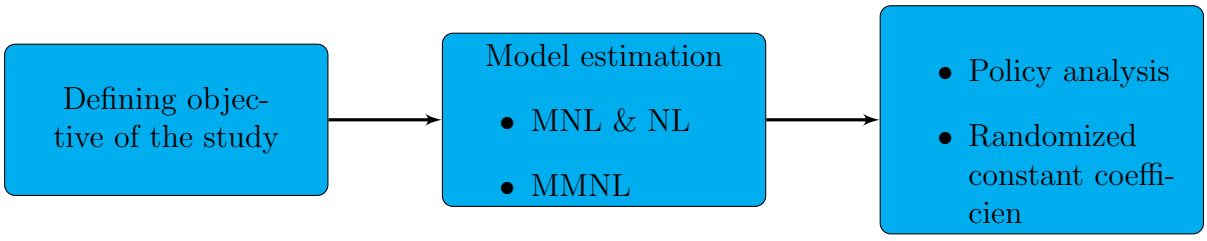


Figure 5.1: Methodology of our study

Chapter 6

Results and discussion

6.1 MNL result

The utility equation for the multinomial logit model can be given by -

$U_{q,Train} = V_{q,Train} + \epsilon_{q,Train}$

$V_{q,Train} = \beta_{train} + \beta_{\text{travel time}} \times$ train travel time for person q + $\beta_{\text{travel cost}} \times$ train travel cost for person q + $\beta_{\text{low income for train}} \times$ low income + $\beta_{\text{medium age train}} \times$ medium age + $\beta_{\text{young age train}} \times$ young age + $\beta_{\text{luggage train}} \times$ luggage + $\beta_{headway} \times$ train headway + $\beta_{\text{gender train}} \times$ male

$U_{q,Car} = V_{q,Car} + \epsilon_{q,Car}$

$V_{q,Car} = \beta_{car} + \beta_{\text{travel time}} \times$ car travel time for person q + $\beta_{\text{travel cost}} \times$ car travel cost for person q + $\beta_{\text{low income for car}} \times$ low income + $\beta_{\text{medium age car}} \times$ medium age + $\beta_{\text{young age car}} \times$ young age + $\beta_{\text{luggage car}} \times$ luggage + $\beta_{\text{gender car}} \times$ male

$U_{q,\text{Swiss metro}} = V_{q,\text{Swiss metro}} + \epsilon_{q,\text{Swiss metro}}$

$V_{q,\text{Swiss metro}} = \beta_{\text{travel time}} \times$ swiss metro travel time for person q + $\beta_{\text{travel cost}} \times$ swiss metro travel cost for person q + $\beta_{headway} \times$ swiss metro headway $\beta_{seat} \times$ airline seats in swiss metro

Table 6.1: Results of MNL model

Explanatory variables	Train		Car		Swiss metro	
	Estimates	t-stat	Estimates	t-stat	Estimates	t-stat
Constants	0.318	1.69	0.313	2.33	0.000	NA
Travel time	-0.012	-26.18	-0.012	-26.18	-0.012	-26.18
Travel cost	-0.007	-17.79	-0.007	-17.79	-0.007	-17.79
Luggage	0.296	3.06	0.089	1.66	0.000	NA
No luggage	0.000	NA	0.000	NA	0.000	NA
High income	0.000	NA	0.000	NA	0.000	NA
Low income	0.812	7.13	0.068	0.70	0.000	NA
Headway	-0.008	-7.25	-	-	-0.008	-7.25
Seats	-	-	-	-	0.708	6.98
Old age	0.000	NA	0.000	NA	0.000	NA
Medium age	-1.199	-8.53	-0.664	-6.17	0.000	NA
Young age	-1.024	-4.94	-1.011	-4.15	0.000	NA
Gender	-0.551	-5.55	0.205	2.81	0.000	NA
Number of cases		8226	Adjusted ρ^2 value		0.115	
Log liklihood value at convergence					-6107	

6.2 NL results

The utility equation for the nested logit model can be given by -

$U_{q,Train} = V_{q,Train} + \epsilon_{q,Train}$

$V_{q,Train} = \beta_{train} + \beta_{\text{travel time}} \times$ train travel time for person q + $\beta_{\text{travel cost}} \times$ train travel cost for person q + $\beta_{\text{low income for train}} \times$ low income + $\beta_{\text{medium age train}} \times$ medium age + $\beta_{\text{young age train}} \times$ young age + $\beta_{\text{luggage train}} \times$ luggage + $\beta_{headway} \times$ train headway + $\beta_{\text{gender train}} \times$ male

$U_{q,Car} = V_{q,Car} + \epsilon_{q,Car}$

$V_{q,Car} = \beta_{car} + \beta_{\text{travel time}} \times$ car travel time for person q + $\beta_{\text{travel cost}} \times$ car travel cost for person q + $\beta_{\text{low income for car}} \times$ low income + $\beta_{\text{medium age car}} \times$ medium age + $\beta_{\text{young age car}} \times$ young age + $\beta_{\text{luggage car}} \times$ luggage + $\beta_{\text{gender car}} \times$ male

$U_{q,\text{Swiss metro}} = V_{q,\text{Swiss metro}} + \epsilon_{q,\text{Swiss metro}}$

$V_{q,\text{Swiss metro}} = \beta_{\text{travel time}} \times$ swiss metro travel time for person q + $\beta_{\text{travel cost}} \times$ swiss metro travel cost for person q + $\beta_{headway} \times$ swiss metro headway $\beta_{seat} \times$ airline seats in swiss metro

Table 6.2: Results of NL model having SM and TRAIN in nest

Explanatory variables	Train		Car		Swiss metro	
	Estimates	t-stat	Estimates	t-stat	Estimates	t-stat
Constants	0.218	0.87	0.393	2.51	0.000	NA
Travel time	-0.013	-23.63	-0.013	-23.63	-0.013	-23.63
Travel cost	-0.007	-17.67	-0.007	-17.67	-0.007	-17.67
Luggage	0.385	2.67	0.106	1.88	0.000	NA
No luggage	0.000	NA	0.000	NA	0.000	NA
High income	0.000	NA	0.000	NA	0.000	NA
Low income	0.997	4.42	0.102	0.99	0.000	NA
Headway	-0.009	-5.72	-	-	-0.009	-5.72
Seats	-	-	-	-	0.888	4.50
Old age	0.000	NA	0.000	NA	0.000	NA
Medium age	-1.440	-5.03	-0.711	-5.96	0.000	NA
Young age	-1.220	-3.93	-1.048	-4.21	0.000	NA
Gender	-0.671	-4.08	0.178	2.27	0.000	NA
$\theta_{SM-TRAIN}$	1.204	6.09	-	-	1.204	6.09
Number of cases		8226	Adjusted ρ^2 value		0.115	
Log liklihood value at convergence					-6107.26	

Table 6.3: Results of NL model having TRAIN and CAR in nest

Explanatory variables	Train		Car		Swiss metro	
	Estimates	t-stat	Estimates	t-stat	Estimates	t-stat
Constants	0.423	2.41	0.375	2.85	0.000	NA
Travel time	-0.012	-24.54	-0.012	-24.54	-0.012	-24.54
Travel cost	-0.007	-17.10	-0.007	-17.10	-0.007	-17.10
Luggage	0.259	2.96	0.084	1.59	0.000	NA
No luggage	0.000	NA	0.000	NA	0.000	NA
High income	0.000	NA	0.000	NA	0.000	NA
Low income	0.694	6.26	0.084	0.91	0.000	NA
Headway	-0.007	-6.89	-	-	-0.007	-6.89
Seats	-	-	-	-	0.714	7.11
Old age	0.000	NA	0.000	NA	0.000	NA
Medium age	-1.130	-8.56	-0.678	-6.47	0.000	NA
Young age	-1.061	-5.32	-0.977	-4.25	0.000	NA
Gender	-0.450	-4.69	0.177	2.49	0.000	NA
$\theta_{CAR-TRAIN}$	0.830	17.20	0.830	17.20	-	-
Number of cases		8226	Adjusted ρ^2 value		0.115	
Log liklihood value at convergence					-6102.8	

Table 6.4: Results of NL model having SM and CAR in nest

Explanatory variables	Train		Car		Swiss metro	
	Estimates	t-stat	Estimates	t-stat	Estimates	t-stat
Constants	0.573	2.33	0.370	2.30	0.000	NA
Travel time	-0.014	-11.02	-0.014	-11.02	-0.014	-11.02
Travel cost	-0.008	-9.83	-0.008	-9.83	-0.008	-9.83
Luggage	0.270	2.71	0.086	1.35	0.000	NA
No luggage	0.000	NA	0.000	NA	0.000	NA
High income	0.000	NA	0.000	NA	0.000	NA
Low income	0.775	6.56	0.076	0.67	0.000	NA
Headway	-0.009	-7.24	-	-	-0.009	-7.24
Seats	-	-	-	-	0.687	6.11
Old age	0.000	NA	0.000	NA	0.000	NA
Medium age	-1.242	-8.46	-0.780	-5.38	0.000	NA
Young age	-1.107	-5.10	-1.192	-3.89	0.000	NA
Gender	-0.525	-5.16	0.255	2.78	0.000	NA
θ_{SM-CAR}	-	-	1.191	10.05	1.191	10.05
Number of cases		8226	Adjusted ρ^2 value		0.115	
Log liklihood value at convergence					-6106.64	

6.3 Mixed multinomial logit model results

The utility equation for the mixed multinomial logit model is given by -

$$U_{q,Train} = V_{q,Train} + \epsilon_{q,Train}$$

$V_{q,Train} = \beta_{train} + \beta'_{\text{travel time}} \times$ train travel time for person q $+ \beta'_{\text{travel cost}} \times$ train travel cost for person q $+ \beta_{\text{low income for train}} \times$ low income $+ \beta_{\text{medium age train}} \times$ medium age $+ \beta_{\text{young age train}} \times$ young age $+ \beta_{\text{luggage train}} \times$ luggage $+ \beta_{headway} \times$ train headway $+ \beta_{\text{gender train}} \times$ male

$$U_{q,Car} = V_{q,Car} + \epsilon_{q,Car}$$

$V_{q,Car} = \beta_{car} + \beta'_{\text{travel time}} \times$ car travel time for person q $+ \beta'_{\text{travel cost}} \times$ car travel cost for person q $+ \beta_{\text{low income for car}} \times$ low income $+ \beta_{\text{medium age car}} \times$ medium age $+ \beta_{\text{young age car}} \times$ young age $+ \beta_{\text{luggage car}} \times$ luggage $+ \beta_{\text{gender car}} \times$ male

$$U_{q,\text{Swiss metro}} = V_{q,\text{Swiss metro}} + \epsilon_{q,\text{Swiss metro}}$$

$V_{q,\text{Swiss metro}} = \beta'_{\text{travel time}} \times$ swiss metro travel time for person q $+ \beta'_{\text{travel cost}} \times$ swiss metro travel cost for person q $+ \beta_{headway} \times$ swiss metro headway $\beta_{seat} \times$ airline seats in swiss metro

Where, for standard normal distribution,

$$\beta'_{\text{travel time}} = \mu_{\text{travel time}} + \sigma_{\text{travel time}} \times \text{draws of travel time}$$

$$\beta'_{\text{travel cost}} = \mu_{\text{travel cost}} + \sigma_{\text{travel cost}} \times \text{draws of travel cost}$$

Now, draws for travel time and travel cost can also come from different distributions like uniform distribution, log normal distribution, triangular distribution and also both travel and travel cost can have different distributions too. Based on the distributions, the randomness for travel time and travel cost is defined on those particular segments only.

Now, let's see those results in the upcoming sections and the detailed iterative results are attached in the appendix in respective segment which we can see by referring tables A.3, A.4, A.5, A.6, A.7, A.8, A.9, A.10, A.11 and A.10 respectively.

6.3.1 Both travel time and travel cost having normal distribution

We are introducing taste heterogeneity to travel time and travel cost only. Now, for standard normal distribution we can express the random coefficient for travel time and travel cost as -

$$\beta'_{\text{travel time}} = \mu_{\text{travel time}} + \sigma_{\text{travel time}} \times \text{draws of travel time}$$

$$\beta'_{\text{travel cost}} = \mu_{\text{travel cost}} + \sigma_{\text{travel cost}} \times \text{draws of travel cost}$$

Table 6.5: Final results of MMNL model for both travel time and travel cost having normal distribution with 400 halton draws.

Explanatory variables	Train		Car		Swiss metro	
	Estimates	t-stat	Estimates	t-stat	Estimates	t-stat
Constants	0.66	2.30	0.80	4.06	0.00	NA
Luggage	0.26	1.49	0.08	0.41	0.00	NA
No luggage	0.00	NA	0.00	NA	0.00	NA
High income	0.00	NA	0.00	NA	0.00	NA
Low income	1.02	4.93	-0.09	-0.53	0.00	NA
Headway	-0.01	-7.54	-	-	-0.01	-7.54
Seats	-	-	-	-	0.29	2.22
Old age	0.00	NA	0.00	NA	0.00	NA
Medium age	-1.06	-4.44	-1.00	-5.12	0.00	NA
Young age	-1.37	-3.24	-1.84	-4.10	0.00	NA
Gender	-0.57	-3.30	0.44	3.12	0.00	NA
μ_{TT}	-0.05	-24.06	0.08	0.41	0.00	NA
σ_{TT}	0.03	23.01	0.03	23.01	0.03	23.01
μ_{TC}	-0.04	-18.22	-0.04	-18.22	-0.04	-18.22
σ_{TC}	0.05	20.64	0.05	20.64	0.05	20.64
Number of cases		914	Adjusted ρ^2 value		0.322	
Log liklihood value at convergence					-5688.47	

6.3.2 Both travel time and travel cost having uniform distribution

We are introducing taste heterogeneity to travel time and travel cost only. Now, for a uniform distribution we can express the random coefficient for travel time and travel cost as -

$$\beta'_{\text{travel time}} = \mu_{\text{travel time}} + \sigma_{\text{travel time}} \times \text{draws of travel time}$$

$$\beta'_{\text{travel cost}} = \mu_{\text{travel cost}} + \sigma_{\text{travel cost}} \times \text{draws of travel cost}$$

Table 6.6: Final results of MMNL model for both travel time and travel cost having uniform distribution with 400 halton draws.

Explanatory variables	Train		Car		Swiss metro	
	Estimates	t-stat	Estimates	t-stat	Estimates	t-stat
Constants	0.82	2.78	0.79	3.30	0.00	NA
Luggage	0.32	1.96	0.13	1.25	0.00	NA
No luggage	0.00	NA	0.00	NA	0.00	NA
High income	0.00	NA	0.00	NA	0.00	NA
Low income	0.98	5.63	-0.08	-2.36	0.00	NA
Headway	-0.01	-7.58	-	-	-0.01	-7.58
Seats	-	-	-	-	0.24	1.80
Old age	0.00	NA	0.00	NA	0.00	NA
Medium age	-1.16	-5.06	-0.96	-4.47	0.00	NA
Young age	-1.79	-4.37	-1.96	-4.41	0.00	NA
Gender	-0.57	-3.39	0.41	3.13	0.00	NA
μ_{TT}	-0.11	-23.19	0.13	1.25	0.00	NA
σ_{TT}	0.12	22.76	0.12	22.76	0.12	22.76
μ_{TC}	-0.13	-20.44	-0.13	-20.44	-0.13	-20.44
σ_{TC}	0.16	19.91	0.16	19.91	0.16	19.91
Number of cases		914	Adjusted ρ^2 value		0.318	
Log liklihood value at convergence					-4700.5	

6.3.3 Both travel time and travel cost having lognormal distribution

We are introducing taste heterogeneity to travel time and travel cost only. Now, for a lognormal distribution we can express the random coefficient for travel time and travel cost as -

$\beta'_{\text{travel time}}$ = -exp($\mu_{\text{travel time}} + \sigma_{\text{travel time}} \times$ draws of travel time)

$\beta'_{\text{travel cost}}$ = -exp($\mu_{\text{travel cost}} + \sigma_{\text{travel cost}} \times$ draws of travel cost)

Here, we are taking negative of exponential to make the travel time and travel cost co-efficient negative. Since, in general they should be negative.

Table 6.7: Final results of MMNL model for both travel time and travel cost having lognormal distribution with 400 halton draws.

Explanatory variables	Train		Car		Swiss metro	
	Estimates	t-stat	Estimates	t-stat	Estimates	t-stat
Constants	0.68	2.53	0.72	3.08	0.00	NA
Luggage	0.46	4.60	0.07	NaN	0.00	NA
No luggage	0.00	NA	0.00	NA	0.00	NA
High income	0.00	NA	0.00	NA	0.00	NA
Low income	0.94	4.99	-0.02	NaN	0.00	NA
Headway	-0.01	-7.97	-	-	-0.01	-7.97
Seats	-	-	-	-	0.48	3.46
Old age	0.00	NA	0.00	NA	0.00	NA
Medium age	-1.25	-5.35	-0.81	-3.66	0.00	NA
Young age	-2.05	-5.81	-1.68	-4.22	0.00	NA
Gender	-0.61	-3.61	0.30	2.24	0.00	NA
μ_{TT}	-5.69	-52.38	0.07	NaN	0.00	NA
σ_{TT}	4.32	26.84	4.32	26.84	4.32	26.84
μ_{TC}	-7.52	-29.48	-7.52	-29.48	-7.52	-29.48
σ_{TC}	6.57	22.34	6.57	22.34	6.57	22.34
Number of cases		914	Adjusted ρ^2 value		0.295	
Log liklihood value at convergence					-4858	

6.3.4 Travel time having lognormal distribution and travel cost having normal distribution

We are introducing taste heterogeneity to travel time and travel cost only. Now, for travel time having lognormal distribution and travel cost having normal distribution we can express the random coefficient for travel time and travel cost as -

$\beta'_{\text{travel time}}$ = -exp($\mu_{\text{travel time}} + \sigma_{\text{travel time}} \times$ draws of travel time)

$\beta'_{\text{travel cost}} = \mu_{\text{travel cost}} + \sigma_{\text{travel cost}} \times$ draws of travel cost

Here, we are taking negative of exponential to make the travel time co-efficient negative. Since, in general this should be negative.

Table 6.8: Final results of MMNL model for travel time having normal distribution and travel cost having lognormal distribution with 400 halton draws.

Explanatory variables	Train		Car		Swiss metro	
	Estimates	t-stat	Estimates	t-stat	Estimates	t-stat
Constants	0.48	1.92	0.48	2.14	0.00	NA
Luggage	0.22	2.13	-0.09	NaN	0.00	NA
No luggage	0.00	NA	0.00	NA	0.00	NA
High income	0.00	NA	0.00	NA	0.00	NA
Low income	0.98	5.32	-0.07	NaN	0.00	NA
Headway	-0.01	-7.60	-	-	-0.01	-7.60
Seats	-	-	-	-	0.57	4.04
Old age	0.00	NA	0.00	NA	0.00	NA
Medium age	-1.03	-4.01	-0.65	-3.06	0.00	NA
Young age	-1.29	-3.55	-1.46	-4.00	0.00	NA
Gender	-0.47	-2.98	0.50	3.83	0.00	NA
μ_{TT}	-0.04	-23.43	-0.09	NaN	0.00	NA
σ_{TT}	0.03	21.87	0.03	21.87	0.03	21.87
μ_{TC}	-1.31	-14.99	-1.31	-14.99	-1.31	-14.99
σ_{TC}	-6.02	-20.92	-6.02	-20.92	-6.02	-20.92
Number of cases		914	Adjusted ρ^2 value		0.306	
Log liklihood value at convergence					-4783.8	

6.3.5 Travel time having normal distribution and travel cost having lognormal distribution

We are introducing taste heterogeneity to travel time and travel cost only. Now, for travel time having lognormal distribution and travel cost having normal distribution we can express the random coefficient for travel time and travel cost as -

$\beta'_{\text{travel time}} = \mu_{\text{travel time}} + \sigma_{\text{travel time}} \times$ draws of travel time

$\beta'_{\text{travel cost}} =$ -exp($\mu_{\text{travel cost}} + \sigma_{\text{travel cost}} \times$ draws of travel cost)

Here, we are taking negative of exponential to make the travel cost co-efficient negative. Since, in general this should be negative.

Table 6.9: Final results of MMNL model for travel time having lognormal distribution and travel cost having normal distribution with 400 halton draws.

Explanatory variables	Train		Car		Swiss metro	
	Estimates	t-stat	Estimates	t-stat	Estimates	t-stat
Constants	0.91	3.21	1.00	4.37	0.00	NA
Luggage	0.45	3.56	0.18	2.91	0.00	NA
No luggage	0.00	NA	0.00	NA	0.00	NA
High income	0.00	NA	0.00	NA	0.00	NA
Low income	1.03	6.05	-0.04	NaN	0.00	NA
Headway	-0.01	-8.33	-	-	-0.01	-8.33
Seats	-	-	-	-	0.09	NaN
Old age	0.00	NA	0.00	NA	0.00	NA
Medium age	-1.27	-5.53	-1.13	-5.21	0.00	NA
Young age	-2.20	-6.23	-1.80	-4.21	0.00	NA
Gender	-0.67	-4.77	0.26	2.48	0.00	NA
μ_{TT}	-1.56	-20.56	0.18	2.91	0.00	NA
σ_{TT}	-3.98	-26.33	-3.98	-26.33	-3.98	-26.33
μ_{TC}	-0.04	-19.13	-0.04	-19.13	-0.04	-19.13
σ_{TC}	0.05	20.58	0.05	20.58	0.05	20.58
Number of cases		914	Adjusted ρ^2 value		0.307	
Log liklihood value at convergence					-4778.3	

6.4 Finding best distribution

Table 6.10: Finding the best fit distribution for travel time and travel cost.

Distribution	Adjusted ρ^2	Log liklihood at convergence	AIC	BIC
Both travel time and travel cost are having normal distribution	0.322	-4675.1	9386.2	9512.5
Both travel time and travel cost are having uniform distribution	0.318	-4700.46	9436.9	9563.2
Both travel time and travel cost are having log normal distribution	0.295	-4858	9752	9878.3
Travel time normal distribution and travel cost log normal distribution	0.306	-4783.8	9603.6	9729.9
Travel time log normal distribution and travel cost normal distribution	0.307	-4778.3	9592.6	9718.8

Now, as per (Ghosh *et al.* 2013) we will compare these models based on AIC, BIC value and the model with distribution having lowest AIC, BIC value will be the best fit distribution for travel time and travel cost.

Now, as per the values tabulated in the table above i.e. on table 6.10, we can clearly see that best fit distribution for travel time and travel cost is normal distribution as for that not only AIC, BIC's are having least value also the model having best adjusted ρ^2 value and log liklihood value more closer to zero.

But, we will adopt lognormal distribution for both travel time and travel cost as division of two lognormal distribution is another lognormal distribution. On the other hand division of two normal distribution is **Cauchy** like distribution which does not have any defined mean for that there might be more fluctuations on value of time.

6.5 Inferences from the model results

From MNL model : MNL model was run to check the consistency of the variables co-efficients. From the table 6.1, we can see -

- The co-efficients of travel time and travel cost are negative which shows the consistency of the result.
- At 90% confidence interval all the coefficients are statistically significant.
- Later, from the policy analysis as shown in the table 7.1 and 7.2, we can see the validation of IIA property for both cost and headway analysis.

For NL model : NL model was run to capture the correlation among the three alternatives(Train, Car & Swiss-metro). From the table 6.2, 6.3 and 6.4, we can see -

- Only Car and Train belong to the same nest. This nesting structure is feasible, whereas the other two nesting structures are not possible because, for them, $\theta > 1$, which is not allowed.
- Later, in the policy analysis, we can clearly validate the above results, as seen from 7.3 and 7.4. Specifically, if we change any attribute of the Train mode, the Car mode becomes a greater gainer or loser compared to the Swissmetro mode

For MMNL model : Here, we will be drawing inferences from our best fit mixed multinomial model and we have found that travel time and travel cost having log normal distribution is the best. So from table 6.7 we can say -

- All of mean and standard deviation of travel time and travel cost coefficients are statistically significant at 95% confidence interval.
- Hence, all the individuals are having taste heterogeneity towards both travel time and travel cost.
- Young age people are least inclined towards train mode as compared to swiss metro mode ceteris paribus .
- Female commuters are more inclined towards car mode as compared to swiss metro mode ceteris paribus.
- Commuters carrying luggage are more inclined towards train mode as compared to other two modes ceteris paribus.

For MMNL having randomized constants : This model was run to capture the heterogeneity across individual observations. Now from the table 6.11 we can say -

- Both the standard error terms for the constant coefficients for train and car are statistically significant which signifies that the individual's observations are independent and the assumption we make to build MNL and NL model was correct.

6.6 Randomized constant co-efficient model results

Since, we have assumed that all the observations across an individual are independent i.e. there will be some heterogeneity across their observations. To catch that we can either consider intra draws in our model structure but that increases computational complexity (Krüger *et al.* 2019). So, we are adopting another way i.e. to randomize the constant co-efficients for our best fit distribution model for travel time and travel cost. We have already found that for travel time and travel cost the best fit distribution is lognormal and now we will allow our co-efficients to be randomized accordingly and tabulate the final model results. For that the random co-efficient equation can be given by -

β'_{train} = -exp($\mu_{\text{train}} + \sigma_{\text{train}} \times$ draws of constant for train)

β'_{car} = -exp($\mu_{\text{car}} + \sigma_{\text{car}} \times$ draws of constant for car)

Here, we are taking negative of exponential to make the travel time and travel cost co-efficient negative. Since, in general they should be negative.

Table 6.11: Final results of MMNL model for both travel time and travel cost having lognormal distribution for randomized constants with 400 halton draws.

Explanatory variables	Train		Car		Swiss metro	
	Estimates	t-stat	Estimates	t-stat	Estimates	t-stat
$\mu_{constant}$	2.93	13.40	2.86	55.64	0.00	NA
$\sigma_{constant}$	-5.13	-8.07	-3.27	-13.29	0.00	NA
Luggage	0.82	4.45	1.66	5.94	0.00	NA
No luggage	0.00	NA	0.00	NA	0.00	NA
High income	0.00	NA	0.00	NA	0.00	NA
Low income	1.71	4.52	0.87	2.54	0.00	NA
Headway	-0.01	-8.21	-	-	-0.01	-8.21
Seats	-	-	-	-	-0.33	-2.01
Old age	0.00	NA	0.00	NA	0.00	NA
Medium age	0.40	NaN	2.05	7.25	0.00	NA
Young age	-1.08	NaN	5.49	9.37	0.00	NA
Gender	0.11	NaN	2.40	8.37	0.00	NA
μ_{TT}	-4.66	-63.79	-4.66	-63.79	-4.66	-63.79
σ_{TT}	3.20	31.61	3.20	31.61	3.20	31.61
μ_{TC}	-4.90	-38.24	-4.90	-38.24	-4.90	-38.24
σ_{TC}	3.77	24.10	3.77	24.10	3.77	24.10
Number of cases		914	Adjusted ρ^2 value		0.38	
Log liklihood value at convergence					-4268.1	

Chapter 7

Policy analysis

7.1 Elasticity effect

Most transportation congestion management actions attempt to effect a change in mode choice during the peak period by influencing the level-of-service variables. According to our dataset, the majority of individuals prefer the Swiss metro over private cars and regional trains, primarily due to its higher reliability and lower travel time. But it would lead to congestion in the metro network. So, to shift a greater share of demand toward the regional train system and alleviate congestion on the metro network, targeted interventions could be implemented. Specifically, reducing the travel cost and headway of trains would enhance their attractiveness. These improvements would reduce waiting time and travel cost of the travelers.
Here, taking a reference from (Bhat 2000) and (Abdelwahab 1998) we first calculated the probabilities of different modes using the cleaned dataset. Then we reduced the travel cost of the train by 2%, 5%, and 10%. Using the new dataset with reduced travel costs, we again calculated the probabilities for each mode and determined the market share by taking the average probability of each mode. Finally, we computed the change in market share for each mode due to the decrease in the travel cost of the train. The change in market share for the train represents self-elasticity, while the changes for the car and Swiss Metro represent cross-elasticity. We have also decreased both the travel cost and headway of the train by certain percentages for the MNL and NL models, as listed in the tables 7.1,7.2, 7.3 and 7.4.

Table 7.1: Effect of decreased travel cost in MNL model

Change in travel cost of train	change in the market share of train	change in the market share of car	change in the market share of SM
decreased by 2%	1.04	-0.08	-0.08
decreased by 5%	2.64	-0.20	-0.19
decreased by 10%	5.37	-0.41	-0.39

Table 7.2: Effect of decreased headway in MNL model

Change in the headway of the train	change in the market share of train	change in the market share of car	change in the market share of SM
decreased by 2%	0.87	-0.06	-0.06
decreased by 5%	2.18	-0.16	-0.16
decreased by 10%	4.45	-0.32	-0.33

Table 7.3: Effect of decreased travel cost in NL model

Change in travel cost of train	change in the market share of train	change in the market share of car	change in the market share of SM
decreased by 2%	1.22	-0.10	-0.07
decreased by 5%	3.09	-0.24	-0.18
decreased by 10%	6.32	-0.49	-0.36

Table 7.4: Effect of decreased headway in NL model

Change in the headway of the train	change in the market share of train	change in the market share of car	change in the market share of SM
decreased by 2%	0.90	-0.07	-0.05
decreased by 5%	2.27	-0.17	-0.13
decreased by 10%	4.60	-0.35	-0.27

Therefore, from the table we can see that for MNL model, on reducing the travel cost and headway of the train mode, there is a increase in the market share of train mode and likewise decrease in the market share of car mode as well as Swiss metro. By observation itself we can see that the decrease in the market share of the other alternatives have reduced by the same percentage, hence it satisfies the IIA property. As we know that, there is no change in the travel cost of car mode and Swiss metro so, there won't be any difference in the change of their market shares as well in the MNL model and we can see this consistency in the results of the table.
But in the case of NL model, there is an unobserved correlation in between train and car with significant θ value of 0.83. So, on decreasing the travel cost or headway of train we can observe the relative decrease in the utility of car mode is more than that of Swiss metro mode. Hence, it doesn't satisfies IIA property.

7.2 Willingness to pay (WTP)

This represents the monetary value that individuals can give to improve the service attributes, like reduced travel time, increased frequency, or seat availability. WTP allows policymakers to quantify how much travelers value different improvements, which can lead to more cost-effective and user friendly improvements. From this willingness to pay value we will get to know that on which service attribute the travelers are more sensitive to, helping to identify which segments of the population are more price-sensitive or time-sensitive. For this analysis we have taken reference from (Ghosh *et al.* 2013) and (Sillano & de Dios Ortúzar 2005).
We have three type of distribution in mixed logit model that are normal distribution, uniform distribution and lognormal distribution, from which we will try to get willingness to pay value for travel time attribute. As the ratio of normal to normal distribution is Cauchy-like distribution and normal to normal distribution gives well-defined but non-standard distribution with distinct properties, but lognormal to lognormal distribution gives lognormal distribution only. So, WTP for lognormal distribution:

$\beta_{TT} = (\mu_{TT}, \sigma_{TT})$ = (-5.688, 4.315)
$\beta_{TC} = (\mu_{TC}, \sigma_{TC})$ = (-7.518, 6.568)

So,
μ_{WTP} = -5.688-(-7.518)=1.83
$\sigma^2_{\text{WTP}} = 4.315^2 + 6.568^2 = 61.758$

$$\mu_{WTP} = e^{\left(\mu_{WTP} + \frac{\sigma^2_{WTP}}{2}\right)} = 1.6 \times 10^{14}$$

Therefore, the value of WTP is 1.6 $\times 10^{14}$ Swiss franc per mins, which is an unrealistic value, so we cannot take the value of time from this distribution. This extreme value arises because the lognormal distribution's heavy left tail allows unbounded values, even when the mean appears reasonable. A plot of distribution of value of time (VOT) with lognormal heterogeneity for VOT vs Frequency is shown in fig.7.1

To mitigate it we can:
1. Restrict the distribution's spread,
2. Use alternative distributions (like, triangular distribution, truncated normal),
3. Report the median WTP instead of the mean, as it is less sensitive to tail effects.

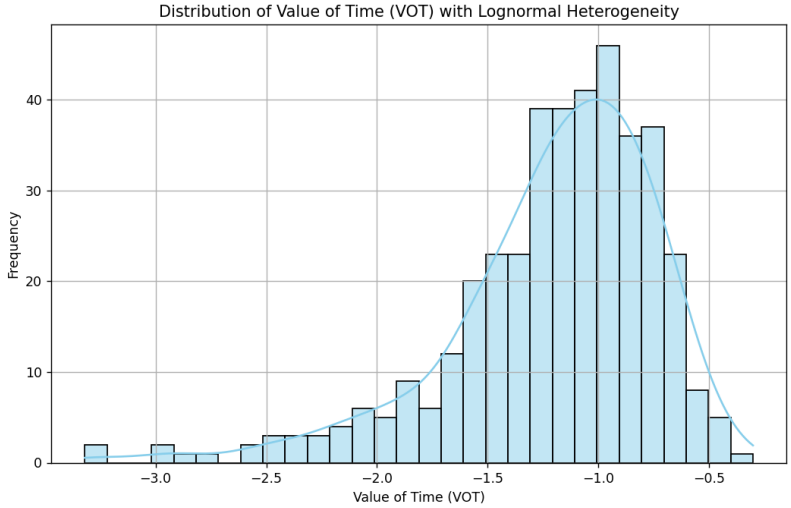


Figure 7.1: Distribution of Value of Time (VOT) with lognormal heterogeneity

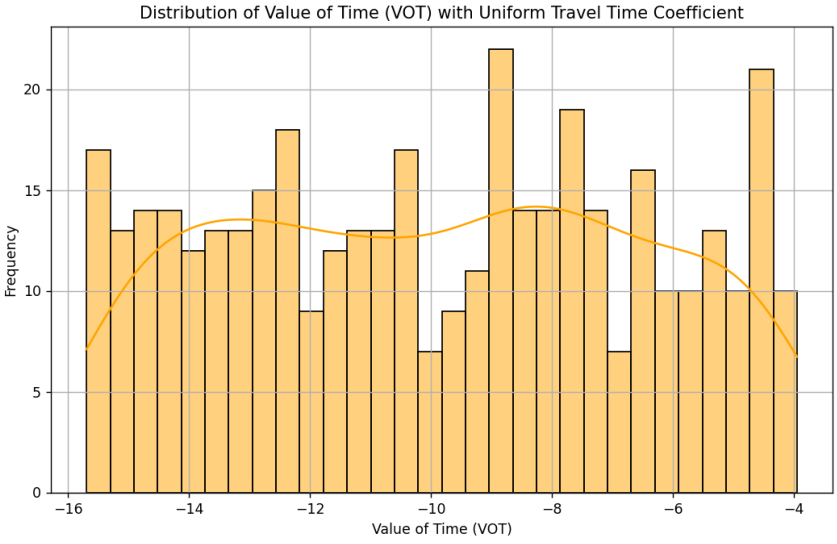


Figure 7.2: Distribution of Value of Time (VOT) with uniform heterogeneity

Now, we can go for uniform distribution to get WTP value for travel time. So, using Monte-Carlo Simulation over,

$\beta_{TT} = (\mu_{TT}, \sigma_{TT})$ = (-0.108, 0.119)
$\beta_{TC} = (\mu_{TC}, \sigma_{TC})$=(-0.127, 0.163)

with 400 halton draws, we get:

Valid draws = 102/400,
mean WTP = 0.8319,
median WTP = 0.8333,
at 95% confidence interval = (0.6634, 1.0588)

So, travelers are willing to pay 0.83 Swiss Franc to save 1 min of travel time.A plot of distribution of Value of Time (VOT) with uniform heterogeneity for VOT vs Frequency on fixing the travel cost coefficient to -0.127 is shown in fig.7.2

This approach ensures that interventions not only shift mode choice behavior but also reflect what users truly value, supporting more efficient and user-centered transportation planning.

Chapter 8

Limitations and future scope of the study

This paper is not the first to study heterogeneity effects in discrete choice modeling or to analyze the Swiss Metro dataset. Previous research on the same dataset has examined similar data using standard choice models like (Belgiawan *et al.* 2017) which compared MNL with an alternative modeling approach called RRM (Random Regret maximization), (Xiao & Zhang 2022), which proposed a non random subset MNL method to analyse travel mode choice and (Antolín *et al.* 2017) which used Discrete-Continuous Maximum Likelihood (DCML) for Nested Logit (NL) model to simultaneously select the optimal nesting structure and estimate parameters
However, this study contributes by explicitly modeling taste heterogeneity in travel time (TT) and travel cost (TC) through different distributional assumptions and by computing willingness-to-pay (WTP) measures, thereby offering deeper behavioral insights.

While traditional discrete choice models like MNL and MMNL require pre-specification of heterogeneity structures, modern methods, such as neural networks (NNs), offer the potential to automatically learn complex patterns from data without manual intervention (Hagenauer & Helbich 2017). However, this comes with a trade-off: although NNs may improve predictive accuracy, they substantially reduce model interpretability. Future research can explore hybrid frameworks that retain the behavioral transparency of discrete choice models while leveraging the flexibility and learning capacity of machine learning techniques.

Apart from that intensive interactions could also have been done to find out the sensitivity of different segments of the population. In mixed multinomial model to find out the best fit distribution, triangular distribution should have been considered. To capture more realistic individual taste heterogeneity across their observations intra draws could have been used.

Chapter 9

Conclusions

This study explored the mode choice behavior of individuals in Switzerland regarding the hypothetical Swiss Metro (SM), train, and car, using advanced discrete choice models to capture heterogeneity and improve policy insights. The research employed three key models—Multinomial Logit (MNL), Nested Logit (NL), and Mixed Multinomial Logit (MMNL)—to analyze the dataset and address the limitations of traditional approaches.

The MNL model provided a baseline understanding of mode preferences, confirming the negative impact of travel time and cost on utility, as expected. However, its inherent Independence of Irrelevant Alternatives (IIA) property limited its ability to capture correlations between alternatives. The NL model addressed this by nesting train and car, revealing stronger substitution patterns between these modes compared to SM, which aligns with their shared unobserved characteristics.

The MMNL model, incorporating random taste heterogeneity, emerged as the most robust framework. By testing various distributions for travel time and cost, the study identified the log-normal distribution as the best fit, despite its computational challenges. This model highlighted significant heterogeneity in preferences, particularly among different age and income groups. For instance, younger individuals showed a stronger aversion to train travel, while low-income travelers were more sensitive to cost changes.

Policy simulations demonstrated that reducing train travel costs or headways could effectively shift demand from SM to trains, alleviating potential congestion in the metro network. The willingness-to-pay (WTP) analysis, though yielding an unrealistic value under log-normal assumptions, underscored the importance of distributional choices in such calculations. A uniform distribution provided a more plausible WTP estimate of 0.83 Swiss Francs per minute of travel time saved.

Despite its contributions, this study has limitations. The reliance on stated preference data may introduce biases, and the assumption of independent observations across individuals could oversimplify real-world decision-making. Future research could explore hybrid models combining machine learning with discrete choice frameworks to enhance predictive accuracy while retaining interpretability. Additionally, incorporating revealed preference data or dynamic choice scenarios could further refine the understanding of mode choice behavior.

In conclusion, this study advances transportation mode choice modeling by addressing heterogeneity and offering policy insights, demonstrating the value of advanced econometric techniques for future research.

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Appendix A

List of Tables

A-1 For MMNL different distribution draws

A-1.1 Co-variance matrix

Table A.1: Variance co-variance matrix for first half of the explanatory variables

Explanatory variables	High income	Medium income	Low income	Old age	Medium age	Young age	Swiss metro headway	Train headway	Luggage
High income	1.00	-0.75	-0.37	0.04	0.15	-0.17	0.00	-0.01	-0.17
Medium income	-0.75	1.00	-0.34	-0.01	0.10	-0.13	0.00	0.01	0.08
Low income	-0.37	-0.34	1.00	-0.04	-0.35	0.42	-0.01	0.00	0.12
Old age	0.04	-0.01	-0.04	1.00	-0.29	-0.12	0.00	0.01	-0.03
Medium age	0.15	0.10	-0.35	-0.29	1.00	-0.57	0.01	0.01	-0.09
Young age	-0.17	-0.13	0.42	-0.12	-0.57	1.00	-0.01	-0.03	0.11
Swiss metro headway	0.00	0.00	-0.01	0.00	0.01	-0.01	1.00	0.01	-0.01
Train headway	-0.01	0.01	0.00	0.01	0.01	-0.03	0.01	1.00	-0.02
Luggage	-0.17	0.08	0.12	-0.03	-0.09	0.11	-0.01	-0.02	1.00
No luggage	0.17	-0.08	-0.12	0.03	0.09	-0.11	0.01	0.02	-1.00
Male	0.28	-0.08	-0.29	0.08	0.02	-0.07	0.01	0.00	-0.14
Train travel time	0.11	0.00	-0.15	0.08	0.06	-0.08	0.01	0.03	-0.04
Train travel cost	0.18	-0.09	-0.13	0.02	0.07	-0.08	0.00	0.02	0.00
Swiss metro travel time	0.05	0.02	-0.09	0.10	0.03	-0.06	0.01	0.02	-0.01
Swiss metro travel cost	0.18	-0.09	-0.12	0.01	0.06	-0.08	0.00	0.02	0.01
Car travel time	0.15	-0.01	-0.20	0.07	0.06	-0.07	0.00	0.03	-0.08
Car travel cost	0.24	-0.03	-0.29	0.08	0.12	-0.19	0.01	0.03	-0.08
Swiss metro airline seats	-0.11	0.00	0.17	-0.06	-0.06	0.09	0.00	0.00	0.15

Table A.2: Variance co-variance matrix for next half of the explanatory variables

Explanatory variables	No luggage	Male	Train travel time	Train travel cost	Swiss metro travel time	Swiss metro travel cost	Car travel time	Car travel cost	Swiss metro airline seats
High income	0.17	0.28	0.11	0.18	0.05	0.18	0.15	0.24	-0.11
Medium income	-0.08	-0.08	0.00	-0.09	0.02	-0.09	-0.01	-0.03	0.00
Low income	-0.12	-0.29	-0.15	-0.13	-0.09	-0.12	-0.20	-0.29	0.17
Old age	0.03	0.08	0.08	0.02	0.10	0.01	0.07	0.08	-0.06
Medium age	0.09	0.02	0.06	0.07	0.03	0.06	0.06	0.12	-0.06
Young age	-0.11	-0.07	-0.08	-0.08	-0.06	-0.08	-0.07	-0.19	0.09
Swiss metro headway	0.01	0.01	0.01	0.00	0.01	0.00	0.00	0.01	0.00
Train headway	0.02	0.00	0.03	0.02	0.02	0.02	0.03	0.03	0.00
Luggage	-1.00	-0.14	-0.04	0.00	-0.01	0.01	-0.08	-0.08	0.15
No luggage	1.00	0.14	0.04	0.00	0.01	-0.01	0.08	0.08	-0.15
Male	0.14	1.00	0.11	0.08	0.07	0.06	0.15	0.24	-0.22
Train travel time	0.04	0.11	1.00	0.32	0.82	0.28	0.58	0.62	-0.18
Train travel cost	0.00	0.08	0.32	1.00	0.22	0.98	0.28	0.36	-0.05
Swiss metro travel time	0.01	0.07	0.82	0.22	1.00	0.17	0.39	0.39	-0.14
Swiss metro travel cost	-0.01	0.06	0.28	0.98	0.17	1.00	0.25	0.33	-0.03
Car travel time	0.08	0.15	0.58	0.28	0.39	0.25	1.00	0.69	-0.20
Car travel cost	0.08	0.24	0.62	0.36	0.39	0.33	0.69	1.00	-0.25
Swiss metro airline seats	-0.15	-0.22	-0.18	-0.05	-0.14	-0.03	-0.20	-0.25	1.00

A-1.2 For both travel time and travel cost normally distributed

Table A.3: Co-efficients of explanatory variables for both travel time and travel cost having normal distribution

Explanatory variables	First iteration - (50 mlhs draws)						Intermediate iteration - (200 mlhs draws)					
	Train		Swiss Metro		Car		Train		Swiss Metro		Car	
	Estimate	t-stat	Estimate	t-stat	Estimate	t-stat	Estimate	t-stat	Estimate	t-stat	Estimate	t-stat
Constants	0.532	1.64	0.669	2.99	0.000	NA	0.600	1.93	0.785	3.61	0.000	NA
Luggage	0.344	2.25	0.155	1.69	0.000	NA	0.311	1.97	0.091	NaN	0.000	NA
No luggage	0.000	NA	0.000	NA	0.000	NA	0.000	NA	0.000	NA	0.000	NA
High income	0.000	NA	0.000	NA	0.000	NA	0.000	NA	0.000	NA	0.000	NA
Low income	1.072	6.23	-0.087	NaN	0.000	NA	1.066	5.23	-0.138	-0.85	0.000	NA
Headway	-0.011	-7.42	-	-	-0.011	-7.42	-0.011	-7.53	-	-	-0.011	-7.53
Seats	-	-	-	-	0.212	1.95	-	-	-	-	0.288	2.06
Old age	0.000	NA	0.000	NA	0.000	NA	0.000	NA	0.000	NA	0.000	NA
Medium age	-1.076	-4.57	-0.994	-5.13	0.000	NA	-1.004	-3.90	-0.950	-4.37	0.000	NA
Young age	-1.507	-3.58	-1.969	-4.52	0.000	NA	-1.382	-3.36	-1.772	-3.96	0.000	NA
Gender	-0.507	-2.96	0.500	3.76	0.000	NA	-0.573	-3.32	0.396	2.83	0.000	NA
μ_{TT}	-0.042	-24.28	-0.042	-24.28	-0.042	-24.28	-0.045	-24.49	-0.045	-24.49	-0.045	-24.49
σ_{TT}	0.034	24.64	0.034	24.64	0.034	24.64	0.033	24.00	0.033	24.00	0.033	24.00
μ_{TC}	-0.038	-20.14	-0.038	-20.14	-0.038	-20.14	-0.041	-16.62	-0.041	-16.62	-0.041	-16.62
σ_{TC}	0.046	20.38	0.046	20.38	0.046	20.38	0.049	18.99	0.049	18.99	0.049	18.99

Table A.4: Co-efficients of explanatory variables for both travel time and travel cost having normal distribution for final two iterations

Explanatory variables	Second last iteration - (350 halton draws)						Final iteration - (400 halton draws)					
	Train		Swiss Metro		Car		Train		Swiss Metro		Car	
	Estimate	t-stat	Estimate	t-stat	Estimate	t-stat	Estimate	t-stat	Estimate	t-stat	Estimate	t-stat
Constants	0.561	1.47	0.670	2.60	0.000	NA	0.656	2.30	0.795	4.06	0.000	NA
Luggage	0.284	1.57	0.077	0.95	0.000	NA	0.263	1.49	0.075	0.41	0.000	NA
No luggage	0.000	NA	0.000	NA	0.000	NA	0.000	NA	0.000	NA	0.000	NA
High income	0.000	NA	0.000	NA	0.000	NA	0.000	NA	0.000	NA	0.000	NA
Low income	1.066	5.84	-0.035	-2.01	0.000	NA	1.023	4.93	-0.088	-0.53	0.000	NA
Headway	-0.011	-7.48	-	-	-0.011	-7.48	-0.011	-7.54	-	-	-0.011	-7.54
Seats	-	-	-	-	0.297	1.93	-	-	-	-	0.290	2.22
Old age	0.000	NA	0.000	NA	0.000	NA	0.000	NA	0.000	NA	0.000	NA
Medium age	-0.983	-3.35	-0.872	-3.88	0.000	NA	-1.061	-4.44	-1.000	-5.12	0.000	NA
Young age	-1.376	-2.93	-1.781	-3.90	0.000	NA	-1.367	-3.24	-1.841	-4.10	0.000	NA
Gender	-0.565	-3.19	0.433	2.94	0.000	NA	-0.567	-3.30	0.443	3.12	0.000	NA
μ_{TT}	-0.045	-23.88	-0.045	-23.88	-0.045	-23.88	-0.045	-24.06	-0.045	-24.06	-0.045	-24.06
σ_{TT}	0.034	22.49	0.034	22.49	0.034	22.49	0.033	23.01	0.033	23.01	0.033	23.01
μ_{TC}	-0.043	-18.15	-0.043	-18.15	-0.043	-18.15	-0.043	-18.22	-0.043	-18.22	-0.043	-18.22
σ_{TC}	0.048	19.54	0.048	19.54	0.048	19.54	0.048	20.64	0.048	20.64	0.048	20.64
For final iteration	No's of observations			914			Adjusted ρ^2			0.322		
	AIC			9386.2			BIC			9512.5		
	Log liklihood at convergence						-4675.1					

A-1.3 For both travel time and travel cost both uniformly distributed

Table A.5: Co-efficients of explanatory variables for both travel time and travel cost having uniform distribution

Explanatory variables	First iteration - (50 mlhs draws)						Intermediate iteration - (200 mlhs draws)					
	Train		Swiss Metro		Car		Train		Swiss Metro		Car	
	Estimate	t-stat	Estimate	t-stat	Estimate	t-stat	Estimate	t-stat	Estimate	t-stat	Estimate	t-stat
Constants	0.703	2.88	0.707	3.73	0.000	NA	0.758	3.17	0.781	3.47	0.000	NA
Luggage	0.342	2.49	0.159	1.49	0.000	NA	0.335	2.68	0.140	1.55	0.000	NA
No luggage	0.000	NA	0.000	NA	0.000	NA	0.000	NA	0.000	NA	0.000	NA
High income	0.000	NA	0.000	NA	0.000	NA	0.000	NA	0.000	NA	0.000	NA
Low income	1.032	6.09	-0.058	-2.28	0.000	NA	1.030	6.09	-0.072	NaN	0.000	NA
Headway	-0.011	-7.67	-	-	-0.011	-7.67	-0.011	-7.65	-	-	-0.011	-7.65
Seats	-	-	-	-	0.184	1.10	-	-	-	-	0.243	1.74
Old age	0.000	NA	0.000	NA	0.000	NA	0.000	NA	0.000	NA	0.000	NA
Medium age	-1.148	-4.90	-0.933	-4.79	0.000	NA	-1.127	-5.47	-0.927	-4.46	0.000	NA
Young age	-1.805	-4.19	-1.900	-4.43	0.000	NA	-1.788	-4.42	-1.900	-4.26	0.000	NA
Gender	-0.517	-3.50	0.436	3.36	0.000	NA	-0.564	-3.46	0.379	2.79	0.000	NA
μ_{TT}	-0.105	-25.36	-0.105	-25.36	-0.105	-25.36	-0.107	-22.45	-0.107	-22.45	-0.107	-22.45
σ_{TT}	0.119	25.16	0.119	25.16	0.119	25.16	0.119	21.85	0.119	21.85	0.119	21.85
μ_{TC}	-0.124	-22.77	-0.124	-22.77	-0.124	-22.77	-0.125	-19.91	-0.125	-19.91	-0.125	-19.91
σ_{TC}	0.158	22.12	0.158	22.12	0.158	22.12	0.160	19.85	0.160	19.85	0.160	19.85

Table A.6: Co-efficients of explanatory variables for both travel time and travel cost having uniform distribution for final two iterations

Explanatory variables	Second last iteration - (350 halton draws)						Final iteration - (400 halton draws)					
	Train		Swiss Metro		Car		Train		Swiss Metro		Car	
	Estimate	t-stat	Estimate	t-stat	Estimate	t-stat	Estimate	t-stat	Estimate	t-stat	Estimate	t-stat
Constants	0.809	3.35	0.772	3.54	0.000	NA	0.821	2.78	0.792	3.30	0.000	NA
Luggage	0.322	2.33	0.133	0.69	0.000	NA	0.315	1.96	0.128	1.25	0.000	NA
No luggage	0.000	NA	0.000	NA	0.000	NA	0.000	NA	0.000	NA	0.000	NA
High income	0.000	NA	0.000	NA	0.000	NA	0.000	NA	0.000	NA	0.000	NA
Low income	0.990	5.97	-0.064	-1.44	0.000	NA	0.984	5.63	-0.078	-2.36	0.000	NA
Headway	-0.011	-7.63	-	-	-0.011	-7.63	-0.011	-7.58	-	-	-0.011	-7.58
Seats	-	-	-	-	0.227	1.58	-	-	-	-	0.237	1.80
Old age	0.000	NA	0.000	NA	0.000	NA	0.000	NA	0.000	NA	0.000	NA
Medium age	-1.152	-4.63	-0.947	-4.20	0.000	NA	-1.164	-5.06	-0.960	-4.47	0.000	NA
Young age	-1.783	-4.17	-1.931	-4.39	0.000	NA	-1.787	-4.37	-1.957	-4.41	0.000	NA
Gender	-0.566	-3.89	0.418	3.19	0.000	NA	-0.568	-3.39	0.413	3.13	0.000	NA
μ_{TT}	-0.107	-23.64	-0.107	-23.64	-0.107	-23.64	-0.108	-23.19	-0.108	-23.19	-0.108	-23.19
σ_{TT}	0.118	23.30	0.118	23.30	0.118	23.30	0.119	22.76	0.119	22.76	0.119	22.76
μ_{TC}	-0.127	-20.45	-0.127	-20.45	-0.127	-20.45	-0.127	-20.44	-0.127	-20.44	-0.127	-20.44
σ_{TC}	0.164	20.31	0.164	20.31	0.164	20.31	0.163	19.91	0.163	19.91	0.163	19.91
For final iteration	No's of observations			914			Adjusted ρ^2			0.318		
	AIC			9436.9			BIC			9563.2		
	Log liklihood at convergence						-4700.46					

A-1.4 For both travel time and travel cost log normally distributed

Table A.7: Co-efficients of explanatory variables for both travel time and travel cost having log normal distribution

Explanatory variables	First iteration - (50 mlhs draws)						Intermediate iteration - (250 mlhs draws)					
	Train		Swiss Metro		Car		Train		Swiss Metro		Car	
	Estimate	t-stat	Estimate	t-stat	Estimate	t-stat	Estimate	t-stat	Estimate	t-stat	Estimate	t-stat
Constants	0.605	2.16	0.650	2.82	0.000	NA	0.681	2.26	0.717	3.02	0.000	NA
Luggage	0.501	3.36	0.083	1.15	0.000	NA	0.487	3.43	0.086	1.14	0.000	NA
No luggage	0.000	NA	0.000	NA	0.000	NA	0.000	NA	0.000	NA	0.000	NA
High income	0.000	NA	0.000	NA	0.000	NA	0.000	NA	0.000	NA	0.000	NA
Low income	0.870	4.86	-0.031	NaN	0.000	NA	0.981	5.35	0.005	4.50	0.000	NA
Headway	-0.011	-7.60	-	-	-0.011	-7.60	-0.011	-7.73	-	-	-0.011	-7.73
Seats	-	-	-	-	0.432	3.12	-	-	-	-	0.486	3.41
Old age	0.000	NA	0.000	NA	0.000	NA	0.000	NA	0.000	NA	0.000	NA
Medium age	-1.231	-5.07	-0.779	-3.88	0.000	NA	-1.265	-5.08	-0.806	-3.83	0.000	NA
Young age	-1.945	-4.84	-1.615	-3.92	0.000	NA	-2.004	-4.79	-1.672	-4.37	0.000	NA
Gender	-0.601	-3.72	0.314	2.33	0.000	NA	-0.620	-3.70	0.284	2.28	0.000	NA
μ_{TT}	-5.883	-50.62	-5.883	-50.62	-5.883	-50.62	-5.697	-54.50	-5.697	-54.50	-5.697	-54.50
σ_{TT}	4.565	25.16	4.565	25.16	4.565	25.16	4.327	28.43	4.327	28.43	4.327	28.43
μ_{TC}	-7.433	-33.53	-7.433	-33.53	-7.433	-33.53	-7.637	-28.88	-7.637	-28.88	-7.637	-28.88
σ_{TC}	6.411	25.06	6.411	25.06	6.411	25.06	6.740	21.67	6.740	21.67	6.740	21.67

Table A.8: Co-efficients of explanatory variables for both travel time and travel cost having log normal distribution for final two iterations

Explanatory variables	Second last iteration - (300 halton draws)						Final iteration - (350 halton draws)					
	Train		Swiss Metro		Car		Train		Swiss Metro		Car	
	Estimate	t-stat	Estimate	t-stat	Estimate	t-stat	Estimate	t-stat	Estimate	t-stat	Estimate	t-stat
Constants	0.658	2.02	0.701	2.78	0.000	NA	0.678	2.53	0.721	3.08	0.000	NA
Luggage	0.475	3.13	0.075	0.68	0.000	NA	0.464	4.60	0.067	NaN	0.000	NA
No luggage	0.000	NA	0.000	NA	0.000	NA	0.000	NA	0.000	NA	0.000	NA
High income	0.000	NA	0.000	NA	0.000	NA	0.000	NA	0.000	NA	0.000	NA
Low income	0.963	5.17	-0.014	-0.82	0.000	NA	0.940	4.99	-0.023	NaN	0.000	NA
Headway	-0.011	-7.55	-	-	-0.011	-7.55	-0.011	-7.97	-	-	-0.011	-7.97
Seats	-	-	-	-	0.468	2.99	-	-	-	-	0.480	3.46
Old age	0.000	NA	0.000	NA	0.000	NA	0.000	NA	0.000	NA	0.000	NA
Medium age	-1.218	-4.22	-0.792	-3.18	0.000	NA	-1.246	-5.35	-0.808	-3.66	0.000	NA
Young age	-1.972	-4.26	-1.701	-3.69	0.000	NA	-2.054	-5.81	-1.678	-4.22	0.000	NA
Gender	-0.598	-3.66	0.318	2.29	0.000	NA	-0.607	-3.61	0.298	2.24	0.000	NA
μ_{TT}	-5.660	-50.76	-5.660	-50.76	-5.660	-50.76	-5.688	-52.38	-5.688	-52.38	-5.688	-52.38
σ_{TT}	4.360	26.74	4.360	26.74	4.360	26.74	4.315	26.84	4.315	26.84	4.315	26.84
μ_{TC}	-7.792	-27.05	-7.792	-27.05	-7.792	-27.05	-7.518	-29.48	-7.518	-29.48	-7.518	-29.48
σ_{TC}	6.802	20.72	6.802	20.72	6.802	20.72	6.568	22.34	6.568	22.34	6.568	22.34
For final iteration	No's of observations			914			Adjusted ρ^2			0.295		
	AIC			9752			BIC			9878.3		
	Log liklihood at convergence						-4858					

A-1.5 Travel time normal distribution and travel cost log normal distribution

Table A.9: Co-efficients of explanatory variables for travel time having normal draw and travel cost having log normal draw

Explanatory variables	First iteration - (50 mlhs draws)						Intermediate iteration - (100 mlhs draws)					
	Train		Swiss Metro		Car		Train		Swiss Metro		Car	
	Estimate	t-stat	Estimate	t-stat	Estimate	t-stat	Estimate	t-stat	Estimate	t-stat	Estimate	t-stat
Constants	0.446	2.22	0.450	2.78	0.000	NA	0.599	2.06	0.503	1.84	0.000	NA
Luggage	0.220	1.60	-0.089	-1.41	0.000	NA	0.207	2.15	-0.079	NaN	0.000	NA
No luggage	0.000	NA	0.000	NA	0.000	NA	0.000	NA	0.000	NA	0.000	NA
High income	0.000	NA	0.000	NA	0.000	NA	0.000	NA	0.000	NA	0.000	NA
Low income	1.043	5.87	-0.033	NaN	0.000	NA	0.908	4.79	-0.145	-1.33	0.000	NA
Headway	-0.011	-7.52	-	-	-0.011	-7.52	-0.011	-7.57	-	-	-0.011	-7.57
Seats	-	-	-	-	0.537	4.49	-	-	-	-	0.555	4.15
Old age	0.000	NA	0.000	NA	0.000	NA	0.000	NA	0.000	NA	0.000	NA
Medium age	-1.049	-5.03	-0.681	-3.85	0.000	NA	-1.122	-4.37	-0.669	-2.64	0.000	NA
Young age	-1.537	-4.14	-1.717	-4.49	0.000	NA	-1.379	-3.12	-1.457	-3.25	0.000	NA
Gender	-0.338	-2.20	0.579	5.38	0.000	NA	-0.479	-2.87	0.498	3.89	0.000	NA
μ_{TT}	-0.043	-23.91	-0.043	-23.91	-0.043	-23.91	-0.043	-23.39	-0.043	-23.39	-0.043	-23.39
σ_{TT}	0.033	24.10	0.033	24.10	0.033	24.10	0.030	24.11	0.030	24.11	0.030	24.11
μ_{TC}	-1.378	-14.83	-1.378	-14.83	-1.378	-14.83	-7.387	-30.19	-7.387	-30.19	-7.387	-30.19
σ_{TC}	-5.826	-21.09	-5.826	-21.09	-5.826	-21.09	6.155	20.13	6.155	20.13	6.155	20.13

Table A.10: Co-efficients of explanatory variables for travel time having normal draw and travel cost having log normal draw for final two iterations

Explanatory variables	Second last iteration - (300 mlhs draws)						Final iteration - (350 mlhs draws)					
	Train		Swiss Metro		Car		Train		Swiss Metro		Car	
	Estimate	t-stat	Estimate	t-stat	Estimate	t-stat	Estimate	t-stat	Estimate	t-stat	Estimate	t-stat
Constants	0.448	1.30	0.504	1.20	0.000	NA	0.478	1.92	0.484	2.14	0.000	NA
Luggage	0.254	0.98	-0.089	-0.41	0.000	NA	0.221	2.13	-0.086	NaN	0.000	NA
No luggage	0.000	NA	0.000	NA	0.000	NA	0.000	NA	0.000	NA	0.000	NA
High income	0.000	NA	0.000	NA	0.000	NA	0.000	NA	0.000	NA	0.000	NA
Low income	0.964	5.07	-0.045	-0.36	0.000	NA	0.983	5.32	-0.071	NaN	0.000	NA
Headway	-0.011	-7.28	-	-	-0.011	-7.28	-0.011	-7.60	-	-	-0.011	-7.60
Seats	-	-	-	-	0.582	4.17	-	-	-	-	0.573	4.04
Old age	0.000	NA	0.000	NA	0.000	NA	0.000	NA	0.000	NA	0.000	NA
Medium age	-1.022	-4.15	-0.680	-2.57	0.000	NA	-1.027	-4.01	-0.650	-3.06	0.000	NA
Young age	-1.412	-3.36	-1.536	-3.37	0.000	NA	-1.288	-3.55	-1.463	-4.00	0.000	NA
Gender	-0.471	-2.70	0.502	3.18	0.000	NA	-0.467	-2.98	0.497	3.83	0.000	NA
μ_{TT}	-0.043	-23.28	-0.043	-23.28	-0.043	-23.28	-0.043	-23.43	-0.043	-23.43	-0.043	-23.43
σ_{TT}	0.032	22.74	0.032	22.74	0.032	22.74	0.031	21.87	0.031	21.87	0.031	21.87
μ_{TC}	-1.316	-14.99	-1.316	-14.99	-1.316	-14.99	-1.306	-14.99	-1.306	-14.99	-1.306	-14.99
σ_{TC}	-6.009	-20.52	-6.009	-20.52	-6.009	-20.52	-6.018	-20.92	-6.018	-20.92	-6.018	-20.92
For final iteration	No's of observations			914			Adjusted ρ^2			0.306		
	AIC			9603.6			BIC			9729.9		
	Log liklihood at convergence						-4783.8					

A-1.6 For travel time log normal distributed and travel cost normally distributed

Table A.11: Co-efficients of explanatory variables for travel time having log normal distribution and travel cost having normal distribution

Explanatory variables	First iteration - (50 mlhs draws)						Intermediate iteration - (150 mlhs draws)					
	Train		Swiss Metro		Car		Train		Swiss Metro		Car	
	Estimate	t-stat	Estimate	t-stat	Estimate	t-stat	Estimate	t-stat	Estimate	t-stat	Estimate	t-stat
Constants	0.799	2.95	0.854	3.65	0.000	NA	0.971	3.39	1.045	4.08	0.000	NA
Luggage	0.343	2.95	0.116	0.95	0.000	NA	0.386	2.65	0.113	1.33	0.000	NA
No luggage	0.000	NA	0.000	NA	0.000	NA	0.000	NA	0.000	NA	0.000	NA
High income	0.000	NA	0.000	NA	0.000	NA	0.000	NA	0.000	NA	0.000	NA
Low income	1.170	6.41	0.003	NaN	0.000	NA	1.088	6.10	-0.013	NaN	0.000	NA
Headway	-0.010	-7.78	-	-	-0.010	-7.78	-0.011	-7.67	-	-	-0.011	-7.67
Seats	-	-	-	-	0.019	0.81	-	-	-	-	0.074	5.00
Old age	0.000	NA	0.000	NA	0.000	NA	0.000	NA	0.000	NA	0.000	NA
Medium age	-1.157	-5.61	-0.949	-4.32	0.000	NA	-1.281	-5.52	-1.111	-5.00	0.000	NA
Young age	-2.270	-6.65	-1.732	-4.12	0.000	NA	-2.266	-6.52	-1.892	-4.20	0.000	NA
Gender	-0.654	-4.37	0.258	1.95	0.000	NA	-0.713	-4.73	0.211	1.47	0.000	NA
μ_{TT}	-1.581	-21.24	-1.581	-21.24	-1.581	-21.24	-5.560	-47.53	-5.560	-47.53	-5.560	-47.53
σ_{TT}	-4.004	-24.95	-4.004	-24.95	-4.004	-24.95	4.050	21.75	4.050	21.75	4.050	21.75
μ_{TC}	-0.044	-19.96	-0.044	-19.96	-0.044	-19.96	-0.042	-18.51	-0.042	-18.51	-0.042	-18.51
σ_{TC}	0.048	21.04	0.048	21.04	0.048	21.04	0.054	17.74	0.054	17.74	0.054	17.74

Table A.12: Co-efficients of explanatory variables for travel time having log normal distribution and travel cost having normal distribution for final two iterations

Explanatory variables	Second last iteration - (300 mlhs draws)						Final iteration - (350 mlhs draws)					
	Train		Swiss Metro		Car		Train		Swiss Metro		Car	
	Estimate	t-stat	Estimate	t-stat	Estimate	t-stat	Estimate	t-stat	Estimate	t-stat	Estimate	t-stat
Constants	0.907	3.21	0.997	4.37	0.000	NA	0.897	2.75	1.024	3.28	0.000	NA
Luggage	0.448	3.56	0.179	2.91	0.000	NA	0.406	2.10	0.152	0.87	0.000	NA
No luggage	0.000	NA	0.000	NA	0.000	NA	0.000	NA	0.000	NA	0.000	NA
High income	0.000	NA	0.000	NA	0.000	NA	0.000	NA	0.000	NA	0.000	NA
Low income	1.033	6.05	-0.042	NaN	0.000	NA	1.090	6.22	-0.022	-3.29	0.000	NA
Headway	-0.011	-8.33	-	-	-0.011	-8.33	-0.011	-7.54	-	-	-0.011	-7.54
Seats	-	-	-	-	0.090	NaN	-	-	-	-	0.072	NaN
Old age	0.000	NA	0.000	NA	0.000	NA	0.000	NA	0.000	NA	0.000	NA
Medium age	-1.274	-5.53	-1.132	-5.21	0.000	NA	-1.263	-5.22	-1.160	-4.95	0.000	NA
Young age	-2.200	-6.23	-1.802	-4.21	0.000	NA	-2.277	-6.04	-1.995	-4.42	0.000	NA
Gender	-0.672	-4.77	0.262	2.48	0.000	NA	-0.649	-3.74	0.278	1.56	0.000	NA
μ_{TT}	-1.563	-20.56	-1.563	-20.56	-1.563	-20.56	-1.546	-22.15	-1.546	-22.15	-1.546	-22.15
σ_{TT}	-3.975	-26.33	-3.975	-26.33	-3.975	-26.33	-3.976	-28.38	-3.976	-28.38	-3.976	-28.38
μ_{TC}	-0.043	-19.13	-0.043	-19.13	-0.043	-19.13	-0.041	-18.16	-0.041	-18.16	-0.041	-18.16
σ_{TC}	0.053	20.58	0.053	20.58	0.053	20.58	0.057	19.26	0.057	19.26	0.057	19.26
For final iteration	No's of observations			914			Adjusted ρ^2			0.307		
	AIC			9592.6			BIC			9718.8		
	Log liklihood at convergence						-4778.3					